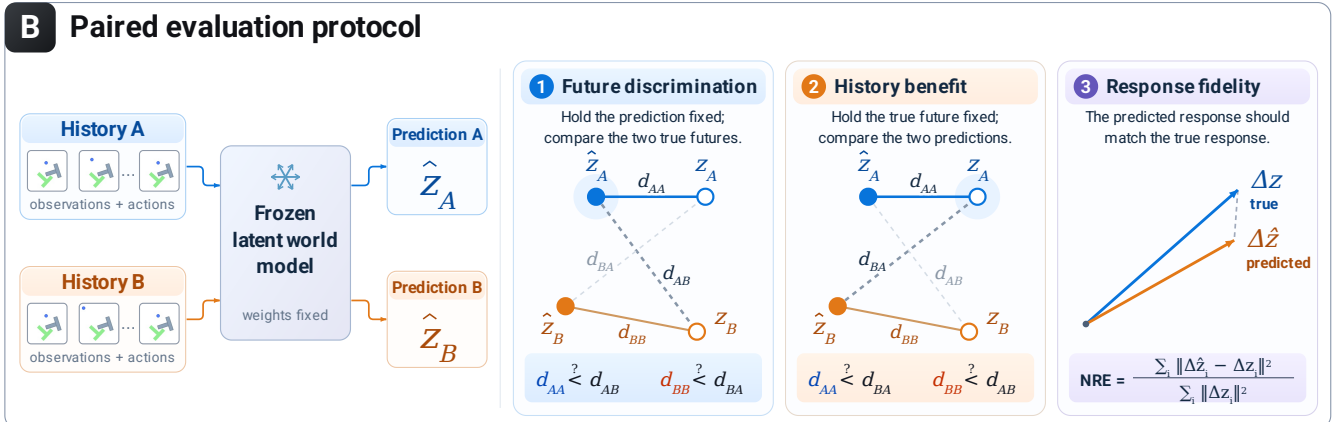

B
Paired evaluation protocol
History A
observations + actions
History B
observations + actions
Frozen latent world model
weights fixed
Prediction A
$\hat{z}_A$
Prediction B
$\hat{z}_B$
1 Future discrimination
Hold the prediction fixed; compare the two true futures.
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
$\hat{z}_B$
z_B
d_{BB}
$d_{AA} \overset{?}{<} d_{AB}$
$d_{BB} \overset{?}{<} d_{BA}$
2 History benefit
Hold the true future fixed; compare the two predictions.
$\hat{z}_A$
z_A
d_{AA}
d_{BA}
d_{AB}
$\hat{z}_B$
z_B
d_{BB}
$d_{AA} \overset{?}{<} d_{BA}$
$d_{BB} \overset{?}{<} d_{AB}$
3 Response fidelity
The predicted response should match the true response.
Δz
true
$\Delta \hat{z}$
predicted
NRE = $\frac{\sum_i \|\Delta \hat{z}_i - \Delta z_i\|^2}{\sum_i \|\Delta z_i\|^2}$